

UNIVERSITY OF CAPE TOWN
Investment Decision Theory and Application
ECO4102Z /ECO4122Z
Activity 6

1. Ms Vuyo Kenke is a portfolio manager at ABC Investments. For all of her clients, Ms Kenke manages portfolios that lie on the Markowitz efficient frontier. Ms Kenke asks Odwa Phakathi, (EAFM, CFA) a managing director at ABC Investments, to review the portfolios of two of her clients, the Table Mountain Manufacturing Company and the Rondebosch Insurance Co. The expected returns of the two portfolios are substantially different. Phakathi determines that the Rondebosch portfolio is virtually identical to the market portfolio and concludes that the Rondebosch portfolio must be superior to the Table Mountain portfolio.

(a) Do you agree or disagree with Phakathi's conclusion that the Rondebosch portfolio is superior to the Table Mountain portfolio? Justify your response with reference to the capital market line. [3 marks]

(b) Ms Kenke remarks that the Rondebosch portfolio has a higher expected return because it has greater nonsystematic risk than Table Mountain's portfolio. Define nonsystematic risk and explain why you agree or disagree with Ms Kenke's remark.

[3 marks]

(c) Ms Kenke is now evaluating the expected performance of two common stocks, AB Bank Ltd and Testing Laboratories Ltd. She has gathered the following information:

- The risk-free rate is 7.5%.
- The expected return on the market portfolio is 14.5%.
- The beta of AB Bank Ltd stock is 1.54.
- The beta of Testing Laboratories stock is 0.85.

Based on her own analysis, Ms Kenke's forecasts of the returns on the two stocks are 16.25% for AB Bank Ltd stock and 13.25% for Testing Laboratories stock. Calculate the required rate of return for AB Bank Ltd stock and for Testing Laboratories stock. Indicate whether each stock is undervalued, fairly valued, or overvalued. [3 marks]

[Total : 9 marks]

2. Download the monthly historical share prices for Compagnie Financière Richemont (Richemont), MTN Group Limited (MTN), Standard Bank Group Limited (Standard Bank), British American Tobacco PLC (BAT), from the JSE, and the All Share Index (Alsi) (keep this data you will use it in the next topic and for test and exam as well) over the period over the period March 2020 to 31 March 2024 and compute the following:
- (a) The correlation coefficient between the returns of each company and the All Share index. [2 marks]
 - (b) The standard deviation of the returns of each company and the index. [2 marks]
 - (c) Alpha for each stock [2 marks]
 - (d) Using Blume's technique, where $\beta_{i2} = 1/3 \times 1 + 2/3 \beta_{i1}$, calculate β_{i2} for the stocks [2 marks]
 - (e) Explain why we need to make the above adjustment. [2 marks]
 - (f) Based on your estimations above, determine whether each of the companies is over or undervalued. [2 marks]
- [Total: 12 marks]
3. One key conclusion of the Capital Asset Pricing Model is that the value of an asset should be measured by considering both the risk and the expected return of the asset, assuming that the asset is held in a well-diversified portfolio. The risk of the asset held in isolation is not relevant under the CAPM.
- (a) True
 - (b) False
4. According to the Capital Asset Pricing Model, investors are primarily concerned with portfolio risk, not the risks of individual stocks held in isolation. Thus, the relevant risk of a stock is the stock's contribution to the riskiness of a well-diversified portfolio.
- a. True
 - b. False
5. If investors become less averse to risk, the slope of the Security Market Line (SML) will increase.
- (a) True
 - (b) False
6. Variance is a measure of the variability of returns, and since it involves squaring the deviation of each actual return from the expected return, it is always larger than its square root, its standard deviation.
- a. True
 - b. False

7. Portfolio A has but one stock, while Portfolio B consists of all stocks that trade in the market, each held in proportion to its market value. Because of its diversification, Portfolio B will by definition be riskless.
- a. True
 - b. False
8. A portfolio's risk is measured by the weighted average of the standard deviations of the securities in the portfolio. It is this aspect of portfolios that allows investors to combine stocks and thus reduce the riskiness of their portfolios.
- a. True
 - b. False
9. The CAPM is built on historic conditions, although in most cases we use expected future data in applying it. Because betas used in the CAPM are calculated using expected future data, they are not subject to changes in future volatility. This is one of the strengths of the CAPM.
- a. True
 - b. False
10. Under the CAPM, the required rate of return on a firm's common stock is determined only by the firm's market risk. If its market risk is known, and if that risk is expected to remain constant, then analysts have all the information they need to calculate the firm's required rate of return.
- a. True
 - b. False
11. A firm can change its beta through managerial decisions, including capital budgeting and capital structure decisions.
- a. True
 - b. False
12. Any change in its beta is likely to affect the required rate of return on a stock, which implies that a change in beta will likely have an impact on the stock's price, other things held constant.
- a. True
 - b. False
13. The slope of the SML is determined by the value of beta.
- a. True
 - b. False

Multiple choice

14. Ruwan's portfolio consists of R700,000 invested in a stock that has a beta of 1.2 and R300,000 invested in a stock that has a beta of 0.8. The risk-free rate is 6% and the market risk premium is 5%. Which of the following statements is CORRECT?
- a. The required return on the market is 10%.
 - b. The portfolio's required return is less than 11%.
 - c. If the risk-free rate remains unchanged but the market risk premium increases by 2%, Ruwan's portfolio's required return will increase by more than 2%.
 - d. If the market risk premium remains unchanged but expected inflation increases by 2%, Ruwan's portfolio's required return will increase by more than 2%.
 - e. If the stock market is efficient, Ruwan's portfolio's expected return should equal the expected return on the market, which is 11%.

Long questions

15. The market price of a security is \$50. Its expected rate of return is 14%. The risk-free rate is 6% and the market risk premium is 8.5%. What will be the market price of the security if its correlation coefficient with the market portfolio doubles (and all other variables remain unchanged)? Assume that the stock is expected to pay a constant dividend in perpetuity.
16. Kaskin, Inc., stock has a beta of 1.2 and Quinn, Inc., stock has a beta of .6. Which of the following statements is most accurate?
- a. The expected rate of return will be higher for the stock of Kaskin, Inc., than that of Quinn, Inc.
 - b. The stock of Kaskin, Inc., has more total risk than Quinn, Inc.
 - c. The stock of Quinn, Inc., has more systematic risk than that of Kaskin, Inc.
17. Two investment advisers are comparing performance. One averaged a 19% rate of return and the other a 16% rate of return. However, the beta of the first investor was 1.5, whereas that of the second was 1.
- a. Can you tell which investor was a better selector of individual stocks (aside from the issue of general movements in the market)?
 - b. If the T-bill rate were 6% and the market return during the period were 14%, which investor would be the superior stock selector?
 - c. What if the T-bill rate were 3% and the market return were 15%?

18. Suppose the rate of return on short-term government securities (perceived to be risk-free) is about 5%. Suppose also that the expected rate of return required by the market for a portfolio with a beta of 1 is 12%. According to the capital asset pricing model:

- a. What is the expected rate of return on the market portfolio?
- b. What would be the expected rate of return on a stock with $\beta = 0.5$?
- c. Suppose you consider buying a share of stock at \$40. The stock is expected to pay \$3 dividends next year and you expect it to sell then for \$41. The stock risk has been evaluated at $\beta = -0.5$. Is the stock overpriced or underpriced?

19. The below results are from a regression of Meta's excess returns over a 60 month period, against the excess returns from the S&P500.

Dependent Variable: Excess Return of META

Independent Variable: Excess Return of S&P 500

Parameter	Estimate	Std. Error	t-Statistic	p-Value
Alpha (α)	0.015	0.003	5.00	0.000
Beta (β)	1.25	0.10	12.50	0.000

Regression Statistics:

- R-squared: 0.65
- Adjusted R-squared: 0.64
- F-statistic: 156.25
- p-value (F-statistic): 0.000
- Number of Observations: 60

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- a. Would you choose to invest in Meta or not? Justify your answer
 - b. Based on the above results, comment on the suitability of CAPM as an asset pricing model.